

CalmSense — Samsung Health Sensor SDK Data Flow

Panic-attack detection from heart rate & HRV · all sensor data processed on-device

Galaxy Watch — Samsung Health Sensor SDK

Tracker: **HEART_RATE_CONTINUOUS** → heart rate + inter-beat interval (IBI). Only this tracker is licensed.



on-device · in-process

CalmSense watch app (wear)

Computes heart rate and HRV (RMSSD from IBI) locally. No raw sensor data is stored or transmitted off the watch.



Wearable Data Layer · encrypted, paired devices only

CalmSense phone app

user consent

On-device classifier flags early signs of a panic attack (elevated HR + suppressed HRV) and distinguishes them from exercise. Triggers breathing guidance & journaling. Explicit in-app consent is obtained at onboarding before any tracking begins.



optional · the user's own account only

Backend (FastAPI) — per-user model only

Stores the user's own derived features (HR, HRV, motion, labels) to retrain that user's personal detection model. No Samsung raw sensor data is shared with third parties.

Data handling: Samsung Health sensor data (HR + IBI) is read and processed **on-device** solely for the individual user's health monitoring and panic-attack support. The user grants explicit consent before tracking starts and can stop monitoring at any time. Derived features stay within the user's own account; raw sensor streams never leave the device.